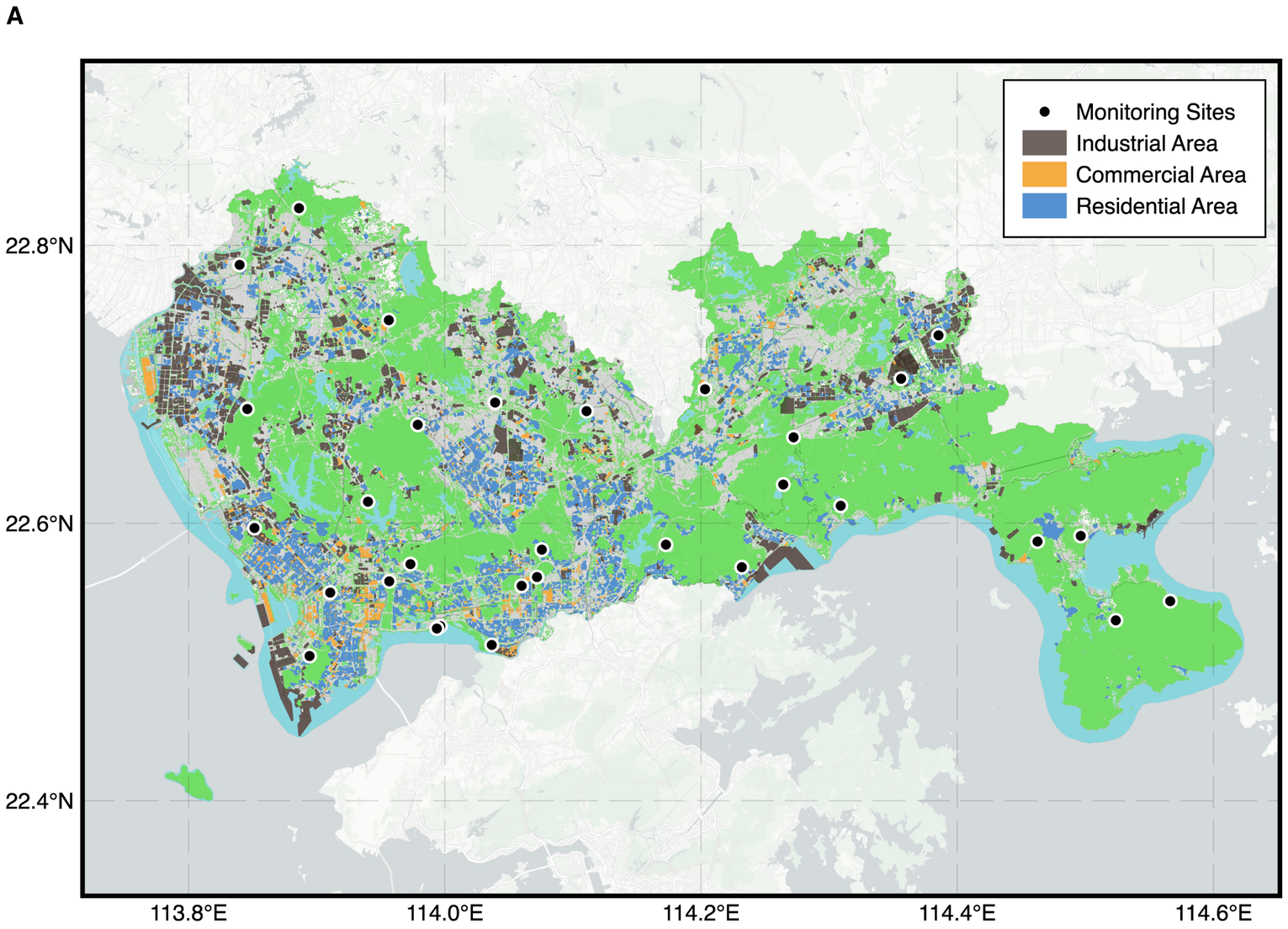
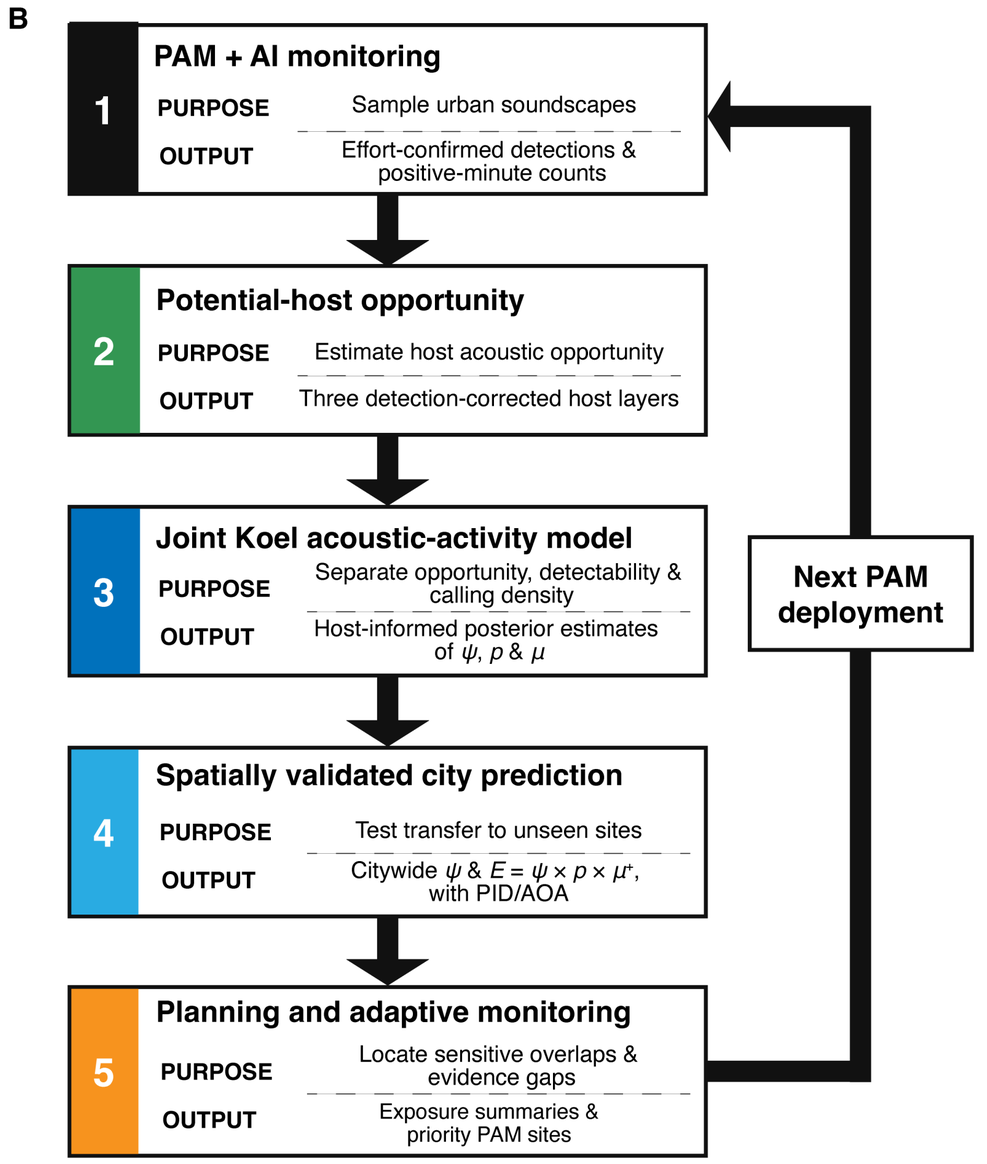

A
Monitoring Sites
Industrial Area
Commercial Area
Residential Area
22.8°N
22.6°N
22.4°N
113.8°E
114.0°E
114.2°E
114.4°E
114.6°E


B
1
PAM + AI monitoring
PURPOSE
Sample urban soundscapes
OUTPUT
Effort-confirmed detections & positive-minute counts
2
Potential-host opportunity
PURPOSE
Estimate host acoustic opportunity
OUTPUT
Three detection-corrected host layers
3
Joint Koel acoustic-activity model
PURPOSE
Separate opportunity, detectability & calling density
OUTPUT
Host-informed posterior estimates of ψ, p & μ
4
Spatially validated city prediction
PURPOSE
Test transfer to unseen sites
OUTPUT
Citywide ψ & E = ψ × p × μ⁺, with PID/AOA
5
Planning and adaptive monitoring
PURPOSE
Locate sensitive overlaps & evidence gaps
OUTPUT
Exposure summaries & priority PAM sites
Next PAM deployment